$$\text{or } (3\lambda-2)\begin{vmatrix}1 & 1 & 1\\ 0 & 3\lambda-11 & 0\\ 0 & 0 & 3\lambda-11\end{vmatrix} = 0 \text{ (by applying } R_2 \to R_2-3R_1, R_3 \to R_3-3R_1)$$

$$\text{or } (3\lambda-2)(3\lambda-11)^2\, 0 \text{ . So } \lambda=\frac{2}{3} \text{ and } \lambda=\frac{11}{3}.$$

We now give an application of system of linear homogeneous equations to chemistry. You are already aware of balancing chemical reaction equations by inspecting the number of atoms present on both sides. A direct method is explained as given below.

Example 1.39

By using Gaussian elimination method, balance the chemical reaction equation:

$C_5H_8 + O_2 \to CO_2 + H_2O.$

(The above is the reaction that is taking place in the burning of organic compound called isoprene.)

Solution

We are searching for positive integers x_1, x_2, x_3 and x_4 such that

$x_1C_5H_8 + x_2O_2 = x_3CO_2 + x_4H_2O$ . .. (1)

The number of carbon atoms on the left-hand side of (1) should be equal to the number of carbon atoms on the right-hand side of (1). So we get a linear homogenous equation

$5x_1 = x_3 \Rightarrow 5x_1 - x_3 = 0$ . ... (2)

Similarly, considering hydrogen and oxygen atoms, we get respectively,

$8x_1 = 2x_4 \Rightarrow 4x_1 - x_4 = 0$, ... (3)

$2x_2 = 2x_3 + x_4 \Rightarrow 2x_2 - 2x_3 - x_4 = 0$. ... (4)

Equations (2), (3), and (4) constitute a homogeneous system of linear equations in four unknowns.

The augmented matrix is $[A \mid B] = \left[\begin{array}{cccc|c} 5 & 0 & -1 & 0 & 0\\ 4 & 0 & 0 & -1 & 0\\ 0 & 2 & -2 & -1 & 0\end{array}\right]$.

By Gaussian elimination method, we get

$$[A \mid B] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc|c} 4 & 0 & 0 & -1 & 0\\ 5 & 0 & -1 & 0 & 0\\ 0 & 2 & -2 & -1 & 0\end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 4 & 0 & 0 & -1 & 0\\ 0 & 2 & -2 & -1 & 0\\ 5 & 0 & -1 & 0 & 0\end{array}\right]$$

$$\xrightarrow{R_3 \to 4R_3 - 5R_1} \left[\begin{array}{cccc|c} 4 & 0 & 0 & -1 & 0\\ 0 & 2 & -2 & -1 & 0\\ 0 & 0 & -4 & 5 & 0\end{array}\right].$$

Therefore, $\rho(A) = \rho([A \mid B]) = 3 < 4 =$ Number of unknowns.

The system is consistent and has infinite number of solutions.

Writing the equations using the echelon form, we get $4x_1 - x_4 = 0, 2x_2 - 2x_3 - x_4 = 0, -4x_3 + 5x_4 = 0$.

So, one of the unknowns should be chosen arbitrarily as a non-zero real number.

Let us choose $x_4 = t, t \neq 0$. Then, by back substitution, we get $x_3 = \frac{5t}{4}, x_2 = \frac{7t}{4}, x_1 = \frac{t}{4}$.

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